From Transistors to Qubits

A Hybrid Learning Framework for Understanding Quantum Computing through Semiconductor Concepts

Course Context: EDC Assignment 2 | **Date:** October 2026
Tech Stack: Python 3, Streamlit, Qiskit 1.x, Qiskit Aer, NumPy, Matplotlib, SVG

Executive Summary & Abstract

Students possessing foundational knowledge of semiconductor devices and classical digital circuits often experience a substantial conceptual gap when transitioning to quantum computing. Classical computing education grounds information in deterministic voltage rails (0 V vs 5 V), macroscopic transistor switches, and non-destructive measurement. Conversely, quantum mechanics introduces non-orthogonal vector superpositions, complex probability amplitudes, unitary transformations, irreversible projective collapse, and quantum entanglement.

This report presents the design, mathematical modeling, and evaluation of **TransistorToQubit** — an interactive software and pedagogical framework. The platform provides a direct, level-by-level bridge from bipolar junction transistor (NPN BJT) switching and canonical logic gates (NOT, AND, OR, NAND, NOR, XOR) to quantum states ($|0\angle$, $|1\angle$, $|+\angle$), quantum logic gates (X, H, Z, CNOT), 3D Bloch sphere geometry, and Aer measurement simulations. Furthermore, a deep comparative analysis of classical additive noise versus quantum decoherence and the 3-qubit bit-flip Quantum Error Correction (QEC) code is formulated and evaluated.

1. Pedagogical Problem & 4-Level Progression Framework

1.1 The Classical-to-Quantum Learning Chasm

Engineering curricula traditionally educate students in a bottom-up physical hierarchy:

$$\text{Semiconductor Materials} \longrightarrow \text{BJT / MOSFET Switches} \longrightarrow \text{Logic Gates} \longrightarrow \text{Digital Systems}$$

In this paradigm, information is represented as a macroscopic physical voltage $V(t)$. Measuring this voltage with a high-impedance oscilloscope probe does not perturb the circuit state.

When entering quantum computing, students encounter axioms that run directly counter to their electronic intuition:

1. **Continuous State Space:** A classical bit is strictly $b \in \{0, 1\}$. A qubit state $|\psi\angle = \alpha\,|0\angle + \beta\,|1\angle$ lives on the continuous surface of a complex unit sphere (the Bloch sphere), where $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$.
2. **Linear Unitary Gates:** Quantum gates must be reversible and unitary ($U^\dagger U = I$), contrasting with classical gates (e.g., AND, OR) that destroy information.
3. **Observer Effect:** Measuring a qubit is not a benign voltage observation; it is a projective measurement that irrevocably collapses $|\psi\angle$ into a computational basis state according to the Born rule ($P(i) = |\langle i \mid \psi\rangle|^2$).

1.2 The Four-Level Progression Framework

To eliminate this cognitive barrier, the framework organizes learning into four structured stages:

- **Level 1 — Transistor Switch:** The student investigates how base current I_B drives an NPN BJT between **Cutoff** ($V_{in} \le V_{th} \Longrightarrow V_{out} \approx 5$ V, Logic 1) and **Saturation** ($V_{in} > V_{th} \Longrightarrow V_{out} \approx 0.2$ V, Logic 0). This grounds binary logic in solid-state semiconductor physics.

- **Level 2 — Logic Gate (NOT ↔ Pauli-X):** The classical BJT inverter circuit ($0 \to 1, 1 \to 0$) is linked to the Pauli-X gate ($X|0\angle = |1\angle$, $X|1\angle = |0\angle$). The platform provides an in-app navigation button taking the student directly from the classical NOT truth table to the quantum X gate in Module 3.
- **Level 3 — Bit to Qubit (Superposition):** The discrete scalar $b \in \{0, 1\}$ transitions to probability amplitudes α, β. The student experiments with the Hadamard gate:

$$H|0\angle = \frac{|0\angle + |1\angle}{\sqrt{2}} \Longrightarrow P(0) = 50\%, \quad P(1) = 50\%$$

- **Level 4 — Classical Circuit vs Quantum Circuit:** The student contrasts deterministic classical gate pipelines against quantum circuits that manipulate interference and entanglement prior to irreversible projective readout.

2. Classical Electronics to Quantum Computing Concept Map

The platform implements the following comprehensive 10-point concept dictionary:

Classical Concept	What Student Knows	Quantum Counterpart	Key Conceptual Distinction
Transistor	Electronic switching device	Qubit (2-level quantum system)	Switch is macroscopic; qubit is a 2-dimensional complex vector space.
ON / OFF	Binary rails (0 V / 5 V)	Measurement collapse (0 or 1)	Classical state is always definite; qubit is continuous until measured.
Voltage/Current	Physical observable signal	Quantum statevector $\|\psi\angle$ / amplitudes	Amplitudes $\alpha, \beta \in \mathbb{C}$ possess phase; voltage is purely real scalar.
Logic Gate	Deterministic Boolean gate	Unitary gate (X, H, Z, CNOT)	Quantum gates must be reversible unitary matrices ($U^\dagger U = I$).
NOT Gate	Inverts signal ($0 \leftrightarrow 1$)	Pauli-X gate (Quantum NOT)	$X\|0\angle = \|1\angle$, $X\|1\angle = \|0\angle$, but acts linearly on superpositions.
Digital Circuit	Interconnected Boolean gates	Quantum Circuit	Exploits superposition and interference across Hilbert space.
Classical Bit	Scalar $b \in \{0, 1\}$	Qubit $\|\psi\angle = \alpha\ \|0\angle + \beta\ \|1\angle$	Continuous continuum of states on the Bloch sphere surface.
Measurement	Non-destructive voltmeter probe	Projective measurement collapse	Irreversible wave-function collapse governed by Born rule ($P = \|\alpha\|^2$).
Noise	Additive voltage $V + n(t)$	Decoherence / Quantum noise	Quantum noise rotates state vectors and destroys phase coherence.

Determinism	Same input yields same output	Probabilistic measurement	Unitary evolution is deterministic; measurement sampling is stochastic.

3. System Architecture & Technical Implementation

3.1 Module 1: Transistor as an Electronic Switch

Module 1 models an NPN BJT Common-Emitter inverter with a resistive collector pull-up:

- **Base Input Voltage (V_{in}):** Synchronized slider and numeric input from $0.0\ \text{V}$ to $6.0\ \text{V}$.
- **Switching Threshold ($\frac{V_{th}}{V_{BE(on)}}$):** Nominal $0.7\ \text{V}$ for standard silicon PN junctions.
- **Collector Load Resistance (R_L):** Variable resistance from 100Ω to $10\{,\}000\Omega$.
- **Current Rating ($I_{C,max}$):** Maximum collector current limit ($1.0\ \text{mA}$ to $50.0\ \text{mA}$).
- **Supply Rail (V_{CC}):** Hardcoded to $5.0\ \text{V}$, representing the universal TTL standard rail.

Mathematical Physics Model: The base current I_B is governed by:

$$I_B = \begin{cases} 0 & \text{if } V_{in} \le V_{th} \\ \frac{V_{in} - V_{th}}{R_B} & \text{if } V_{in} > V_{th} \end{cases}$$

where $R_B = 10\ \text{k}\Omega$. In the active region, $I_{C,\text{active}} = \beta I_B$ ($b\eta = 100$). The maximum current allowed by the DC load line is:

$$I_{C,\text{sat}} = \frac{V_{CC} - V_{CE,\text{sat}}}{R_L} \quad (V_{CE,\text{sat}} = 0.2\ \text{V})$$

The actual collector current is $I_C = \min\left(I_{C,\text{active}}, I_{C,\text{sat}}, I_{C,max}\right)$.

- **Cutoff ($V_{in} \le V_{th}$):** $I_C = 0\ \text{mA}$, $V_{out} = V_{CC} = 5.0\ \text{V} \Longrightarrow \textbf{Logic 1}$ (Open switch).
- **Saturation ($I_{C,\text{active}} \ge I_{C,\text{sat}}$):** $I_C = I_{C,\text{sat}}$, $V_{out} = V_{CE,\text{sat}} \approx 0.2\ \text{V} \Longrightarrow \textbf{Logic 0}$ (Closed switch).

Live visual feedback is rendered through an in-memory dynamic SVG schematic illustrating wire conduction, base activation, collector current flow, and digital output state.

3.2 Module 2: Classical Logic Gates

Module 2 provides interactive exploration of the six canonical gates required by the assignment document:

- **Gates Implemented:** NOT, AND, OR, NAND, NOR, and XOR.
- **Dual Visualizations:** Students toggle between standard IEEE 91-1984 logic symbols and Resistor-Transistor Logic (RTL) transistor-level circuit schematics.
- **Dynamic Truth Tables:** Rendered inside an isolated HTML frame with real-time row highlighting based on user switch inputs ($A, B \in \{0, 1\}$).
- **NOT $\rightarrow$ X Navigation:** Selecting the NOT gate exposes an interactive card and one-click button:

  $$\text{NOT } (0) = 1 \Longleftrightarrow X|0\angle = |1\angle$$

  Clicking the button seamlessly routes the session to the Pauli-X gate in Module 3.

3.3 Module 3: Quantum Gates & Circuits (Qiskit Engine)

Module 3 is built upon Qiskit 1.x and Qiskit Aer:

- **Initial Computational States:** Selectable ground state $|0\angle = \begin{pmatrix}1\\0\end{pmatrix}$ and excited state $|1\angle = \begin{pmatrix}0\\1\end{pmatrix}$.
- **Quantum Operators:**

  $$X = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \quad H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}, \quad Z = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$$

- **Two-Qubit Entanglement (CNOT):** Applies Controlled-NOT:

$$\text{CNOT}\ (H|0\angle \otimes |0\angle) = \frac{|00\angle + |11\angle}{\sqrt{2}} = |\Phi^+\angle$$

- **Annotated 3D Bloch Sphere:** Features cardinal coordinate axes and labeled Dirac points ($|0\angle$ North Pole, $|1\angle$ South Pole, $|+\angle, |-\angle, |i\angle$ along the equatorial plane).
- **Measurement Collapse Simulation:** Executes projective measurements on `AerSimulator` across 128 to 4096 shots, producing empirical probability histograms matching theoretical Born rule calculations ($P(m) = |\langle m \mid \psi\rangle|^2$).

3.4 Module D: Comparison & Noise Framework

Module D unifies the learning experience by providing:

1. The 10-row Classical $\leftrightarrow$ Quantum Concept Mapping table.
2. The 4-Level Learning Progression with contextual navigation buttons.
3. The Classical vs Quantum Noise comparison suite.

4. Deep-Dive: Classical Noise vs Quantum Decoherence

4.1 Classical Noise & Voltage Noise Margins

In classical digital electronics, noise is additive voltage fluctuations:

$$V_{out}(t) = V_{\text{ideal}} + n(t), \quad n(t) \sim \mathcal{N}(0, \sigma^2)$$

Digital circuits achieve immense reliability through **noise margins**:

$$NM_H = V_{OH,min} - V_{IH,min}, \quad NM_L = V_{IL,max} - V_{OL,max}$$

Because the BJT inverter has a steep, non-linear voltage transfer characteristic, any small fluctuation ($\sigma < 1.0$ V) is attenuated at subsequent stages. Information is refreshed automatically.

4.2 Quantum Decoherence & Error Channels

In quantum systems, noise is non-additive and phase-sensitive:

1. **Bit-Flip Error Channel (X):** $|\psi\angle \xrightarrow{p} X|\psi\angle$, randomly flipping $|0\angle \leftrightarrow |1\angle$.
2. **Phase-Flip Error Channel (Z):** $|\psi\angle \xrightarrow{p} Z|\psi\angle$, altering phase without affecting populations ($|+\angle \leftrightarrow |-\angle$).
3. **Depolarizing Noise Channel ($\mathcal{E}$):** Replaces the density matrix with the maximally mixed state with probability p:

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$$

4.3 Quantum Error Correction (3-Qubit Bit-Flip Code)

Classical error correction uses simple duplication ($0 \to 000, 1 \to 111$). In quantum computing, the **No-Cloning Theorem** ($U|\psi\angle\,|0\angle \neq |\psi\angle\,|\psi\angle$) forbids copying an unknown state. Furthermore, directly measuring qubits to detect errors collapses their superposition.

The 3-qubit bit-flip code resolves this via entanglement:

1. **Encoding:** Using two CNOT gates, the logical qubit is encoded into an entangled 3-qubit state:

$$|\psi\angle = \alpha\ |0\angle + \beta\ |1\angle \implies |\psi_{\text{enc}}\angle = \alpha\ |000\angle + \beta\ |111\angle$$

2. **Syndrome Measurement:** Parity operators Z_0Z_1 and Z_1Z_2 are measured using ancilla qubits, yielding syndrome bits $s_1s_2 \in \{00, 10, 11, 01\}$ without revealing α or β:
   - $s_1s_2 = 00 \implies$ No error
   - $s_1s_2 = 10 \implies$ Bit-flip on q_0 $(\to$ apply $X_0)$
   - $s_1s_2 = 11 \implies$ Bit-flip on q_1 $(\to$ apply $X_1)$
   - $s_1s_2 = 01 \implies$ Bit-flip on q_2 $(\to$ apply $X_2)$
3. **Recovery:** Applying the diagnosed Pauli-X operator restores the original state $|\psi\angle$ with 100% theoretical fidelity.

5. Experimental Verification & Results Matrix

Specification / Criterion	Implementation Output	Verification
Vary Base Input Voltage	V_{in} slider + manual numeric input (0.0 V to 6.0 V)	Passed
Vary Switching Threshold	V_{th} slider + numeric input (0.1 V to 4.0 V)	Passed
Vary Load Resistance	R_L slider + numeric input (100Ω to 10 kΩ)	Passed
Vary Collector Current Limit	$I_{C,max}$ slider + numeric input (1.0 mA to 50.0 mA)	Passed
Observe Transistor Switching	Dynamic SVG schematic showing Cut-off vs Saturation paths	Passed
Canonical Logic Gates	NOT, AND, OR, NAND, NOR, XOR with IEEE symbols & RTL circuits	Passed
Live Truth Tables	HTML frame with active-row blue highlight for current inputs	Passed
Quantum Basis States	$\|0\angle$, $\|1\angle$ selectable in Module 3	Passed
Quantum Gates & Circuits	X, H, Z, CNOT, and Measurement with Qiskit 1.x & Aer	Passed
Bloch Sphere Representation	3D Bloch sphere with labeled poles ($\|0\angle, \|1\angle, \|+\angle, \|-\angle, \|i\angle$)	Passed
Classical $\leftrightarrow$ Quantum Comparison	Dedicated Module D with 10-row mapping table	Passed
4-Level Progression	Explicit 4-level sequence with NOT $\rightarrow$ X navigation bridge	Passed
Noise Modeling & QEC	Classical $n(t)$ sim, bit/phase-flip channels, 3-qubit QEC pipeline	Passed

6. Conclusion

The **TransistorToQubit** framework provides a pedagogically sound, mathematically rigorous bridge connecting classical electronic engineering with modern quantum computation. By structuring the learning journey around four progressive levels, anchoring abstract quantum gates in physical transistor switching, and illustrating the fundamental difference between classical noise margins and quantum error correction, the platform substantially accelerates student comprehension of quantum information science.

7. References

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